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EXPERIMENT 1
Regular Expressions in Python

Aim

To write a Python program that demonstrates how to use regular expressions to match and search for patterns in text.

Algorithm

• Import the re module.

• Store a sample sentence in a string variable.

• Use re.search() to find the first occurrence of the word “Python” in the text.

• Use re.findall() to find all occurrences of the word “Python” in the text.

• Use re.match() to check whether the text starts with the word “Python”.

• Display the results of all three operations.

Code

```
import re

text = "Python is easy. Python is powerful."

match = re.search("Python", text)
print("Search:", match.group())

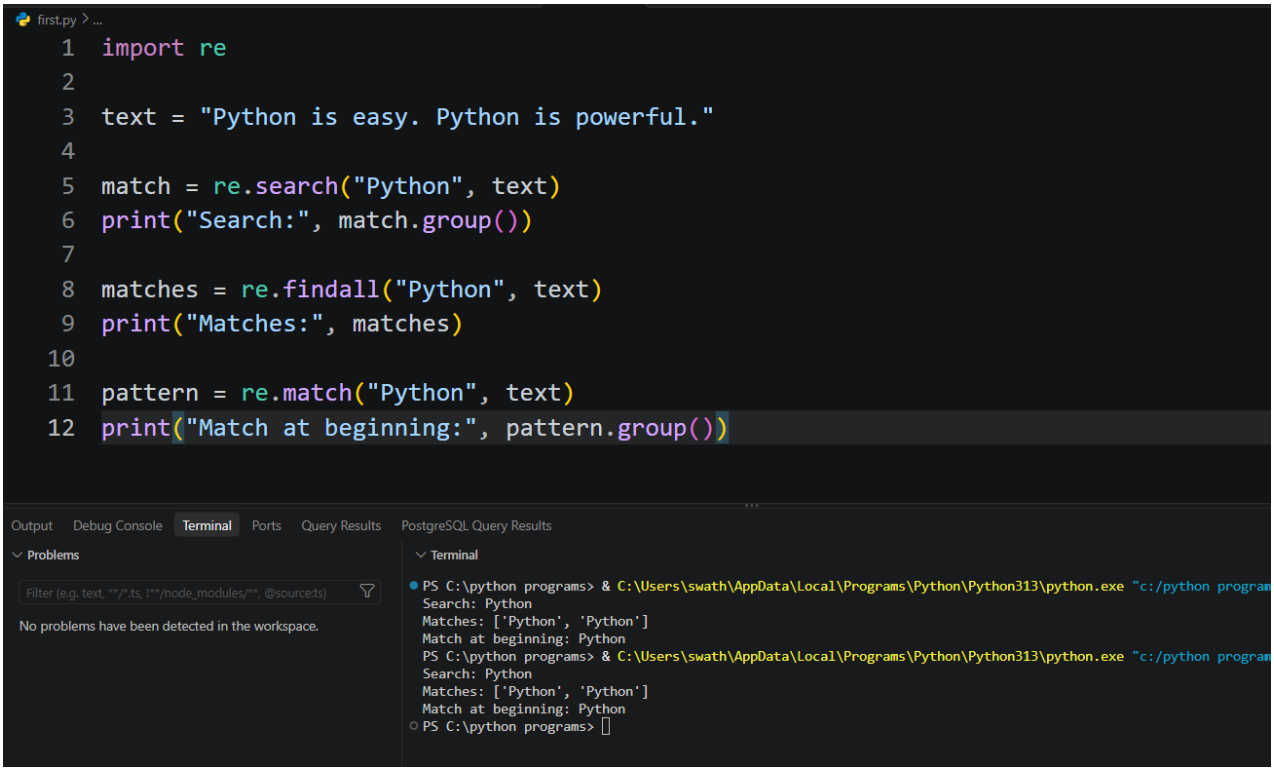
matches = re.findall("Python", text)
print("Matches:", matches)

pattern = re.match("Python", text)
print("Match at beginning:", pattern.group())
```

Output

The following screenshot shows the program execution and output:

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Result

The program was executed successfully. It demonstrates that re.search() finds the first matching occurrence, re.findall() returns all matching occurrences, and re.match() checks for a match at the beginning of the string.